

Test Specifications: Tickets Domain

Gema Event Management Platform
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Source file: routes/ticket.routes.ts
Controllers: ticket.controller.ts
Middleware: authenticate, authorize

Role matrix summary:

- generate: vendor, admin
- generate-missing: user, customer, vendor, admin
- verify-qr: employee, vendor, admin
- checkin: employee, vendor, admin
- event/:eventId (list): employee, vendor, admin
- user/my-tickets: user, customer
- order/:orderId: user, customer
- /:ticketId/download: user, customer
- /:ticketId (details): user, customer, employee, vendor, admin
- /:ticketId/transfer: user, customer
- /:ticketId/resend: user, customer

All routes require authenticate (global router.use).

Ticket Generation

Test ID	Description	Preconditions	Input	Expected Output	Priority
TC-TKT-001	Happy path: vendor generates tickets for an order	Vendor authenticated, order exists and is paid	POST /tickets/generate with valid { orderId }	HTTP 201; tickets created in DB; QR generation queued in BullMQ	High
TC-TKT-002	Admin generates tickets	Admin authenticated, paid order	POST /tickets/generate with valid { orderId }	HTTP 201; tickets created	High
TC-TKT-003	User role blocked from generating tickets	Authenticated as user	POST /tickets/generate	HTTP 403; role not authorized	High
TC-TKT-004	Customer role blocked from generating tickets	Authenticated as customer	POST /tickets/generate	HTTP 403	High
TC-TKT-005	Employee role blocked from generating tickets	Authenticated as employee	POST /tickets/generate	HTTP 403	High
TC-TKT-006	Unauthenticated request to generate	No token	POST /tickets/generate	HTTP 401	High
TC-TKT-007	Duplicate ticket generation for same order	Tickets already generated	POST /tickets/generate with same orderId	HTTP 409 or idempotent 200; no duplicate tickets created	High
TC-TKT-008	Generate for non-existent order	No order with given ID	POST /tickets/generate with invalid orderId	HTTP 404	Medium
TC-TKT-009	Generate missing tickets — user generating for own order	User authenticated, paid order missing tickets	POST /tickets/generate-missing with valid orderId	HTTP 200 or 201; missing tickets created	High
TC-TKT-010	Generate missing tickets — non-existent or other user's order	User authenticated	orderId belonging to different user	HTTP 403 or 404	High

QR Verification & Check-in

Test ID	Description	Preconditions	Input	Expected Output	Priority
TC-TKT-011	Happy path: employee verifies valid QR code	Employee authenticated, ticket exists and is valid	POST /tickets/verify-qr/:eventId with valid QR payload	HTTP 200; ticket details returned; ticket not yet checked in	High
TC-TKT-012	Vendor verifies QR code for own event	Vendor authenticated	POST /tickets/verify-qr/:eventId	HTTP 200	High
TC-TKT-013	Admin verifies QR code	Admin authenticated	POST /tickets/verify-qr (no eventId)	HTTP 200	High
TC-TKT-014	User role blocked from QR verification	Authenticated as user	POST /tickets/verify-qr/:eventId	HTTP 403	High
TC-TKT-015	QR verification — invalid/tampered QR data	Employee authenticated	Malformed or tampered QR payload	HTTP 400; invalid ticket error	High
TC-TKT-016	QR verification — ticket for wrong event	Employee authenticated for Event A	QR belongs to Event B	HTTP 400 or 403; event mismatch error	High
TC-TKT-017	QR verification — already checked-in ticket	Employee authenticated, ticket already checked in	Valid QR of checked-in ticket	HTTP 200 or 409 with alreadyCheckedIn: true in response	High
TC-TKT-018	QR verification — cancelled/refunded ticket	Employee authenticated	QR for refunded ticket	HTTP 400; ticket not valid for entry	High
TC-TKT-019	Happy path: check in a valid ticket	Employee authenticated, ticket valid and not checked in	POST /tickets/:ticketId/checkin	HTTP 200; checkedIn: true, checkInTime set on ticket	High
TC-TKT-020	Check in — already checked in	Employee authenticated	ticketId of already checked-in ticket	HTTP 409; duplicate check-in error	High
TC-TKT-021	Check in — refunded ticket	Employee authenticated	ticketId of refunded ticket	HTTP 400; invalid status for check-in	High
TC-TKT-022	Check in — non-existent ticket	Employee authenticated	Non-existent ticketId	HTTP 404	Medium
TC-TKT-023	Check in — user role blocked	Authenticated as user	POST /tickets/:ticketId/checkin	HTTP 403	High
TC-TKT-024	Check in — invalid ObjectId format	Employee authenticated	ticketId = "notanid"	HTTP 422 or 400; invalid ID format	Medium

Ticket Retrieval

Test ID	Description	Preconditions	Input	Expected Output	Priority
TC-TKT-025	Happy path: get all tickets for an event	Vendor/employee authenticated, event has tickets	GET /tickets/event/:eventId	HTTP 200; paginated list of tickets for that event	High
TC-TKT-026	Get event tickets — user role blocked	Authenticated as user	GET /tickets/event/:eventId	HTTP 403	High
TC-TKT-027	Get event tickets — non-existent event	Vendor authenticated	Non-existent eventId	HTTP 200 with empty array or HTTP 404	Medium
TC-TKT-028	Get event tickets — vendor accessing another vendor's event	Vendor A authenticated	eventId of Vendor B's event	HTTP 403; ownership check fails	High
TC-TKT-029	Happy path: user retrieves own tickets	User/customer authenticated	GET /tickets/user/my-tickets	HTTP 200; list of user's tickets only	High
TC-	Mv-tickets — vendor role blocked	Authenticated as vendor	GET /tickets/user/my-	HTTP 403	High

TKT-030	tickets				
TC-TKT-031	My-tickets — unauthenticated	No token	GET /tickets/user/my-tickets	HTTP 401	High
TC-TKT-032	Happy path: get tickets by order ID	Customer authenticated, owns order	GET /tickets/order/:orderId	HTTP 200; tickets belonging to that order	High
TC-TKT-033	Get tickets by order — other user's order	Customer A authenticated	orderId of Customer B	HTTP 403 or 404	High
TC-TKT-034	Happy path: get single ticket details	Any authenticated role	GET /tickets/:ticketId	HTTP 200; full ticket object	High
TC-TKT-035	Get ticket details — non-existent ticket	Any authenticated role	Non-existent ticketId	HTTP 404	Medium
TC-TKT-036	Get ticket details — unauthenticated	No token	GET /tickets/:ticketId	HTTP 401	High

Ticket Download

Test ID	Description	Preconditions	Input	Expected Output	Priority
TC-TKT-037	Happy path: download ticket PDF	Customer authenticated, owns ticket	GET /tickets/:ticketId/download	HTTP 200; PDF binary or signed Cloudinary URL returned	High
TC-TKT-038	Download PDF — vendor role blocked	Authenticated as vendor	GET /tickets/:ticketId/download	HTTP 403	High
TC-TKT-039	Download PDF — other user's ticket	Customer A authenticated	ticketId belonging to Customer B	HTTP 403	High
TC-TKT-040	Download PDF — non-existent ticket	Customer authenticated	Invalid ticketId	HTTP 404	Medium
TC-TKT-041	Download PDF — unauthenticated	No token	GET /tickets/:ticketId/download	HTTP 401	High

Ticket Transfer

Test ID	Description	Preconditions	Input	Expected Output	Priority
TC-TKT-042	Happy path: transfer ticket to another user	Customer authenticated, owns valid ticket	POST /tickets/:ticketId/transfer with { recipientEmail }	HTTP 200; ticket ownership transferred; both users notified by email	High
TC-TKT-043	Transfer — recipient email does not exist	Valid ticket owner	recipientEmail not in DB	HTTP 404; recipient not found	High
TC-TKT-044	Transfer — already checked-in ticket	Customer owns ticket	ticketId of checked-in ticket	HTTP 400; checked-in tickets cannot be transferred	High
TC-TKT-045	Transfer — refunded ticket	Customer	Refunded ticketId	HTTP 400; invalid status	High
TC-TKT-046	Transfer — other user's ticket	Customer A	ticketId owned by Customer B	HTTP 403	High
TC-TKT-047	Transfer — vendor role blocked	Authenticated as vendor	POST /tickets/:ticketId/transfer	HTTP 403	High
TC-	Transfer — recipient is Customer				

TKT-048	same as sender	authenticated	recipientEmail = own email	HTTP 400; cannot transfer to self	Medium
TC-TKT-049	Transfer — unauthenticated	No token	POST /tickets/:ticketId/transfer	HTTP 401	High

Ticket Resend

Test ID	Description	Preconditions	Input	Expected Output	Priority
TC-TKT-050	Happy path: resend ticket email	Customer authenticated, owns ticket	POST /tickets/:ticketId/resend	HTTP 200; ticket email queued for resend via BullMQ	High
TC-TKT-051	Resend — other user's ticket	Customer A authenticated	ticketId owned by Customer B	HTTP 403	High
TC-TKT-052	Resend — vendor role blocked	Authenticated as vendor	POST /tickets/:ticketId/resend	HTTP 403	High
TC-TKT-053	Resend — non-existent ticket	Customer authenticated	Non-existent ticketId	HTTP 404	Medium
TC-TKT-054	Resend — unauthenticated	No token	POST /tickets/:ticketId/resend	HTTP 401	High
TC-TKT-055	Resend — email queue failure	BullMQ worker down	Valid resend request	HTTP 500 or 202 with warning; error logged; ticket email not sent	Medium

Security Tests

Test ID	Description	Preconditions	Input	Expected Output	Priority
TC-TKT-056	IDOR: customer accesses another customer's ticket	Customer A authenticated	GET /tickets/:ticketId where ticket belongs to Customer B	HTTP 403 or 404	Critical
TC-TKT-057	Role escalation: user attempts QR verification	Authenticated as user	POST /tickets/verify-qr/:eventId	HTTP 403	Critical
TC-TKT-058	Forged QR code check-in attempt	Employee authenticated	QR payload with modified ticketId	HTTP 400; invalid ticket signature	Critical
TC-TKT-059	Employee accesses event they are not assigned to	Employee authenticated	GET /tickets/event/:eventId for unassigned event	HTTP 403	High
TC-TKT-060	Mass assignment: attempt to set checkedIn: true via transfer body	Customer authenticated	POST /tickets/:ticketId/transfer with { checkedIn: true } in body	Ignored; only recipientEmail processed	High